

1 - (3) Quantitative sales forecasting (Three Period Moving Average & Extrapolation)



Do Now

Theme 2 knowledge recall

State the main benefits and drawbacks of sales forecasting?



Learning Outcome

To be able to calculate and interpret three-period moving averages for sales forecasting

To Know



- Explain what time-series analysis is and why it's used in business forecasting.
- Describe the components of a time series (trend, seasonal, cyclical, and random variations).
- Calculate three-period moving averages accurately using sales data.
- Interpret the results of a moving average to identify trends.
- Interpret scatter graphs and draw a line of best fit.
- How to use extrapolation to predict future sales based on past data.

Quantitative sales forecasting

Quantitative forecasting is a data-based mathematical process that sales teams use to understand performance and predict future sales based on historical data and patterns. Forecasting results give businesses the ability to make informed decisions on strategies and processes to ensure continuous success.

Three Period Moving Average

Step 1	Take the first 3 data points and then calculate the average (add up > divide by 3)
Step 2	Plot this average against the middle value of the 3 (see example in red below)
Step 3	Drop the first value, then repeat the process by adding on the next value to calculate the average for the next 3 data points.

Date	Sales	3 year moving average
2012	400	
2013	500	556.6
2014	770	

Practice Question

- Calculate the 3-point moving average
- Calculate the variation for each year
- Calculate the average variation

Year	Sales (£ Millions)	3 Point Moving Average	Variation
2016	14		
2017	16		
2018	18		
2019	21		
2020	19		
2021	21		
2022	24		

Average Variation =



10/22

Extract A

UK pet care market is set to grow

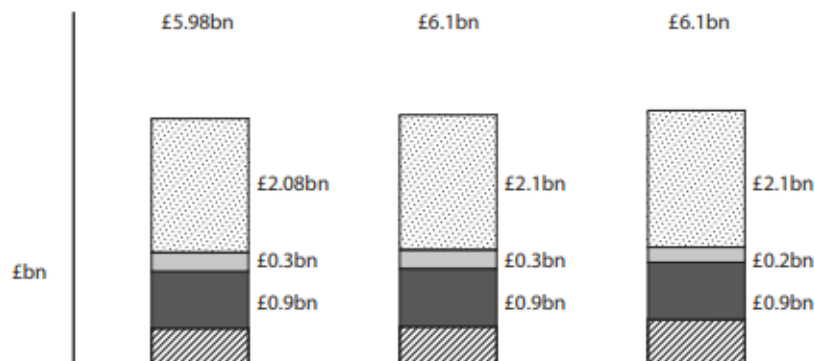
Analysts have predicted steady growth in the UK pet care market. The pet care market includes pet food, pet accessories, pet grooming and veterinary services. The forecast value of this market in 2020–2021 is £7bn. This is due to a large increase in the UK's pet population since the start of the global health crisis.

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Pet accessories such as dog beds have a positive outlook. Sales of pet accessories are expected to grow and exceed £0.9bn by 2021–2022.

Key

- Pet food
- Pet accessories
- Pet grooming
- Veterinary services



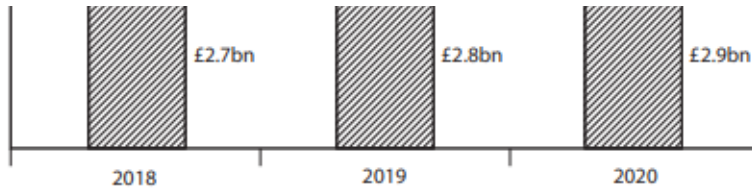


Figure 1: UK pet care market 2018–2020

Extract E

The percentage of UK households owning a pet from 2012–2021

Year	Households owning a pet (%)
2012	47
2013	48
2014	47
2015	46
2016	40
2017	44
2018	45
2019	40
2020	41
2021	60

Using the data in Extract E, calculate the three year moving average for the percentage of UK households who owned a pet in 2020. You are advised to show your working.

(4)

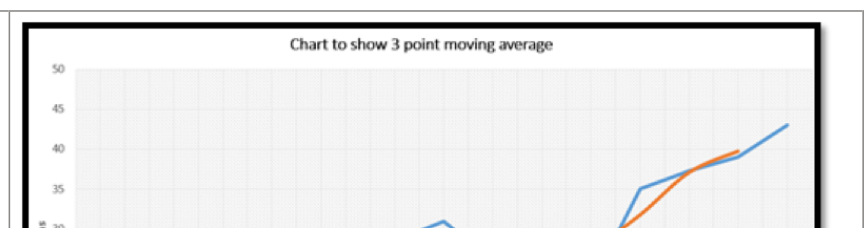
The sales forecasting process

This example demonstrates how to calculate a 3 point moving average, it is important to understand how this can be used to extrapolate future values. The steps below will provide a walked-through example of how this works

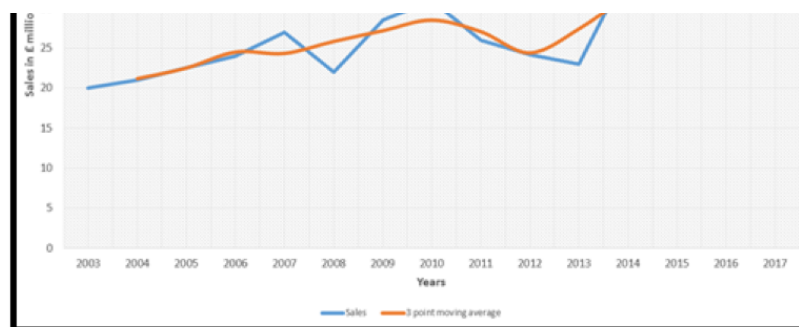
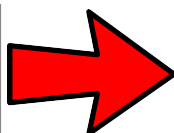
Walkthrough Steps 1 and 2

Calculating the moving average and plotting onto a graph (highly unlikely that you will be asked to add anything to a graph)

Year	Sales (£ millions)	3 point Moving average
2003	20	
2004	21	21.17
2005	22.5	22.5
2006	24	24.5
2007	27	24.33

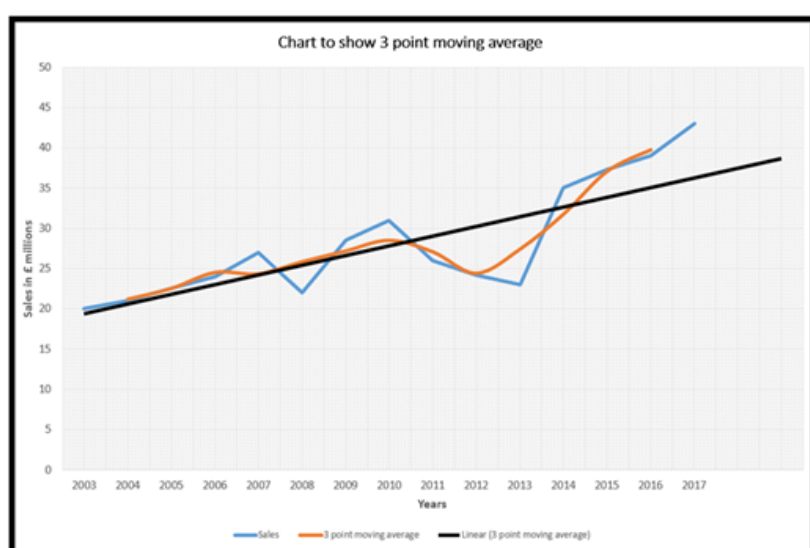


Year	Sales	3 point moving average
2008	22	25.83
2009	28.5	27.17
2010	31	28.5
2011	26	27.07
2012	24.2	24.4
2013	23	27.4
2014	35	31.77
2015	37.3	37.1
2016	39	39.77
2017	43	



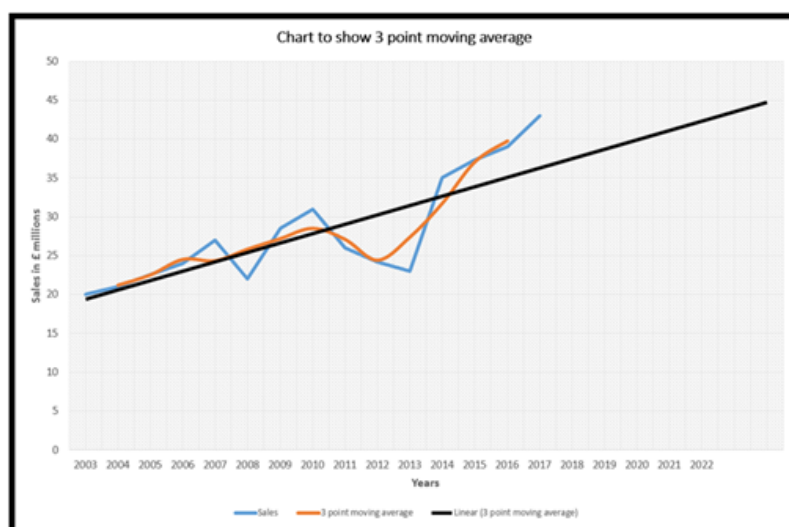
Walkthrough Step 3

Next step is to add a line of best fit onto the graph – this is a straight line through the data with equal values on either side of the line. **Note - you will not need to actually draw one of these in an exam, understanding how one is drawn is enough.**



Walkthrough Step 4

Take the line of best fit and extend it forwards (in a straight line) for the next 5 years beyond 2017 to 2022. You can then read off predicted sales levels on the graph. Here the figure for 2022 is **£42.5 million**.



Walkthrough step 5

Year	Sales (£ millions)	3 point Moving average	Variation
2003	20		
2004	21	21.17	-0.17
2005	22.5	22.5	0
2006	24	24.5	-0.5
2007	27	24.33	2.67
2008	22	25.83	-3.83
2009	28.5	27.17	1.33
2010	31	28.5	2.5
2011	26	27.07	-1.07
2012	24.2	24.4	-0.2
2013	23	27.4	-4.4
2014	35	31.77	3.23
2015	37.3	37.1	0.2
2016	39	39.77	-0.77
2017	43		

1) Go back to the original table and calculate the variation between the original data and the moving average

$$\text{Variation} = \text{Sales} - 3 \text{ point moving average}$$

2) The total variations (when they are all added up) is **-£1.01 million**

3) Divide the total amount of variations -£1.01 by the number of variations **13 = -£0.08 million (rounded up)** - This is the **AVERAGE VARIANCE FROM TREND**

4) Add -£0.08 million to 2022 sales forecast of £42.50 million

5) This gives us a final result of **£42.42 million** - the adjusted sales forecast to take account of the average variance.

Would be very difficult to exam this calculation without doing a lot of it for you.

Let's take the £42.5m extrapolated figure for 2022 (see step 4). Now we need to adjust the figure for variation. This is because the line of best fit doesn't take into account all the peaks and troughs in the lines, so we make another calculation:

Discuss the possible limitations of quantitative sales forecasting for future sales at *Pets at Home*.

(8)

Define Key term:	
Paragraph 1 POINT	One ... is.... This is because...
IMPLICATION/S	Consequently/This means/This will lead to...
WHAT IMPACT DOES THE IMPLICATION/S HAVE ON THE BUSINESS	Therefore...

OBJECTIVE/S?	As a result, ...
Paragraph 2 POINT	In addition another ... is....because....
IMPLICATION/S	Consequently/This means/This will lead to...
WHAT IMPACT DOES THE IMPLICATION/S HAVE ON THE BUSINESS OBJECTIVE/S?	Therefore...
Paragraph 3 POINT	However, ...
	This is because...
IMPLICATION/S	Consequently/This means/This will lead to...
WHAT IMPACT DOES THE IMPLICATION/S HAVE ON THE BUSINESS OBJECTIVE/S?	Therefore...
	As a result, ...